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Department of Computer Science and Engineering

Introduction to Artificial Intelligence and Machine Learning (CSN12601)

Lab Manual: Assignment-5

Lab Title: Programs on NumPy, Math Module and Functions

Objective:

1. To learn and practice operations on NumPy arrays using various methods for manipulation, sorting, and accessing elements.
2. To understand the concepts of assignment, shallow copy, and deep copy in Python, and how they differ in terms of data management.
3. To explore the functionality of Python's math and cmath modules for performing mathematical and complex number operations.
4. To develop problem-solving skills by implementing a calculator using functions, enhancing understanding of function definitions, arguments, and return values in Python.

Materials Required:

- Computer with internet access
- Python (latest version)
- Python Integrated Development Environment (IDLE)
- Access to an online Python environment (e.g., Google Colaboratory)

Theory:

NumPy Arrays and Array Manipulation: NumPy is a powerful library in Python used for numerical computing. It provides support for large, multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on these arrays. Understanding how to manipulate arrays using methods like `append()`, `insert()`, `pop()`, `remove()`, `index()`, `sort()`, and `reverse()` is crucial for data manipulation and analysis tasks in Python.

Array Copying Techniques:

- **Assignment Operator (=):** This creates a reference to the original array, meaning changes to the new array will affect the original array.
- **Shallow Copy (`numpy.copy()`):** This creates a new array object, but the elements themselves are references to the objects found in the original array. Changes in mutable elements will reflect in both copies.
- **Deep Copy (`copy.deepcopy()`):** This creates a completely independent copy of the original array, including all nested objects. Changes in the copied array do not affect the original array.

Math and Cmath Modules:

- **Math Module:** Provides mathematical functions that operate on real numbers, such as `math.acos()`, `math.ceil()`, `math.exp()`, and `math.gcd()`. These functions are useful for performing trigonometric calculations, rounding numbers, computing exponentials, and finding the greatest common divisor, among other operations.
- **Cmath Module:** Designed for complex number operations, the `cmath` module includes functions such as `cmath.tan(x)`, `cmath.log10(x)`, and `cmath.isclose()`, which operate on complex numbers. This module extends the functionality of the math module to handle complex data types.

Creating a Calculator Using Functions: Building a calculator in Python using functions involves defining separate functions for each arithmetic operation, such as addition, subtraction, multiplication, and division. This exercise helps solidify understanding of how functions are defined, how parameters are passed, and how results are returned. It also demonstrates the importance of modularity and reusability in programming, as functions can be reused across different parts of a program or in different projects.

List of Programs in Assignment-5

1) Create a NumPy array and apply methods such as append(), insert(), pop(), remove(), index(), sort(), and reverse().
2) Write a program to copy array with assignment operator, shallow copy method and deep copy method.
3) Write a program for math and cmath module functionality in your program. E.g., math.acos(), math.ceil(), math.exp(), math.gcd(), cmath.tan(x), cmath.log10(x), cmath.isclose().
4) Write a program to create a calculator using function.